

IT5005 Artificial Intelligence

Tutorial 8

1. Consider the following constraint satisfaction problem:

Variables:

$$A, B, C$$

Domains:

$$D_A = D_B = D_C = \{0, 1, 2, 3, 4\}$$

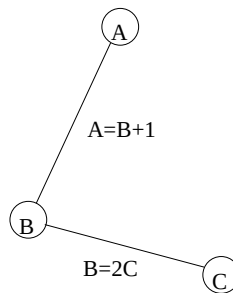
Constraints:

$$A = B + 1 \quad B = 2C$$

Construct a constraint graph for this problem. Show a trace of the AC-3 algorithm on this problem. Assume that initially, the arcs in queue are in the order $\{(A, B), (B, A), (B, C), (C, B)\}$.

Solution:

Constraint Graph:



Original domains:

$$D_A = D_B = D_C = \{0, 1, 2, 3, 4\}$$

Content of queue and domain of variables at the end of each iteration:

Revised Domain	Queue
$D_A = \{0, 1, 2, 3, 4\}$	(A,B) (B,A) (B,C) (C,B)
$D_A = \{1, 2, 3, 4\}$	(B,A) (B,C) (C,B)
$D_B = \{0, 1, 2, 3\}$	(B,C) (C,B)
$D_B = \{0, 2\}$	(C,B) (A,B)
$D_C = \{0, 1\}$	(A,B)
$D_A = \{1, 3\}$	

Allowable domain values:

$$D_A = \{1, 3\}$$

$$D_B = \{0, 2\}$$

$$D_C = \{0, 1\}$$

Note that of course not all of these values are compatible: we need to choose one of them and then the rest will be implied (e.g. $A = 1 \rightarrow B = 0 \rightarrow C = 0$).

2. Consider the 4-queens problem on a 4×4 chess board. Suppose the leftmost column is column 1, and the topmost row is row 1. Let Q_i denote the row number of the queen in column i , $i = 1, 2, 3, 4$. Assume that variables are assigned in the order Q_1, Q_2, Q_3, Q_4 , and the domain values of Q_i are tried in the order 1, 2, 3, 4. Show a trace of the backtracking algorithm with forward checking to solve the 4-queens problem.

Solution:

The following is the trace of the search tree:

